# Aaron Chou

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## EDUCATION

**National Taiwan University (NTU)**, *B.Sc. in Mechanical Engineering* — Taipei, Taiwan

- Last 60 GPA: 4.03/4.3 | CGPA: 3.80/4.3 — Sept. 2018 – Jan. 2023
- **Coursework:** Automatic Control, Digital Control System, Kinematics, Dynamics, Computer Programming

**Aoyama Gakuin University (AGU)**, *Exchange Program* — Tokyo, Japan

- **Coursework:** Data Structures and Algorithms, Introduction to Computer Systems — Sept. 2022 – Jan. 2023

## PUBLICATIONS

[1] **Y.L. Chou**, L.C. Wang, Y.S. Luo, Y.T. Liu, J.L. Li. *VL-N3RD-Bench: Benchmarking Vision-Language Navigation with 3D Gaussian Splatting Reconstruction for Deployment*, ICRA Workshop: VLA Pipelines for Real Robots, 2026. (under review)

[2] Y.S. Luo, L.C. Wang, H. Mandala, **Y.L. Chou**, G.H.G. Christmann, C.Y. Lee, W.C. Chen, et al. *Feasibility-Guided Planning over Multi-Specialized Locomotion Policies*, IEEE Int. Conf. Robot. Autom. (ICRA), 2026. (link)

## EXPERIENCE

**Inventec Corporation**, AI Center, Robotics — Taipei, Taiwan

*Robotics Research Engineer — Quadruped Robots* — Mar. 2025 – Present

- **Vision-Language Navigation (VLN)**: Benchmark sim-to-real VLN by evaluating a pretrained VLA model on quadrupeds across reconstructed and real-world scenes using 3D Gaussian Splatting, analyzing the impact of reconstruction quality on navigation. [1]
- **Learning-Based Control**: Developed a low-level control SDK for quadruped platforms with no official API, enabling deployment (link) of learning-based locomotion policies trained in **IsaacGym** and establishing reusable infrastructure.
- **Terrain-Aware Navigation**: Trained a 15 cm gap-traversal policy, implemented a feasibility-guided planning framework supporting skill selection on hybrid terrain, and contributed to an ICRA 2026 submission [2].
- **SLAM Optimization**: Improved single-LiDAR SLAM by **dual-LiDAR fusion** (link) in ICRA 2025 QRC, achieving a **65%** reduction in mapping-to-navigation time and enabling rapid terrain reconstruction in dynamic environments.

**Flytech Technology Co. Ltd.** — Taipei, Taiwan

*Robotics Engineer — Autonomous Mobile Robots* — Mar. 2024 – Jan. 2025

- Engineered a precise docking system (demo) by integrating motion control, localization, and a customized rack tracker, achieving navigation accuracy of **±1–1.5 cm** to minimize redocking attempts in production deployments.
- Developed and maintained **10+ ROS packages** and delivered a patrol demo showcased at Computex 2024.

**Chien Kuo High School**, Robotics team, FRC#8020 CyberpunK — Taipei, Taiwan

*Youth Mentor — Mechanical Design* — Feb. 2022 – Jul. 2023

- Mentored 30+ students for robot design, CAD/CAM, CNC operation, and testing of the competition-ready robot (demo).

## PROJECTS

**Isaac-MoveIt Manipulator Integration**, *Collaborative Project* (demo) — Jun. 2025 – Present

- Integrated **IsaacSim** with **MoveIt2**, enabling manipulator control in simulation for future dexterous research.

## SKILLS

**Robotics:** IsaacSim/Lab, ROS/ROS2 (Nav2, MoveIt2, FAST-LIO), 3D Reconstruction (3DGS), Sensor Fusion, SLAM

**Hardware & Software:** Unitree Go1/A1, AMR, Livox LiDAR, RealSense, IMU, RTX 5090; Docker, Git, Blender, PCL, SolidWorks

**Programming:** Python, C++, Shell Scripting (Bash), LaTeX

## HONORS & COMPETITIONS

**2025 ICRA Quadruped Robot Challenge (QRC) — Participation Certificate, Autonomous** (demo) — Atlanta, GA

**2022 FIRST Robotics Competition (FRC) Sacramento Regional, Finalist (Team Mentor)** — Sacramento, CA

**2020 Dean's List Award (top 5% of the class in the semester)** — Taipei, Taiwan